

INSPECTION REPORT

Equipment: Centrifugal Pump P-101

Location: Crude Distillation Unit (CDU)

Date: 2026-08-15

Inspector: R. Sharma (MRPL Inspection Division)

Report ID: MRPL-INS-2026-0847

Classification: CONFIDENTIAL · Internal Use Only

INSPECTION FINDINGS

1. MECHANICAL SEAL INSPECTION

- Primary mechanical seal shows visible leakage on the atmospheric side.
- Leak rate estimated at 15-20 drops per minute.
- Seal face wear pattern indicates misalignment.
- Carbon face deposits observed on the seal surface.

2. VIBRATION ANALYSIS

- Overall vibration measured at 12.4 mm/s RMS (alarm threshold: 7.1 mm/s).
- Dominant frequency at 1x running speed (2980 RPM) suggests imbalance.
- Sub-synchronous vibration at 0.43x detected, possible oil whirl.
- Bearing housing temperature: 92°C (normal: <75°C).

3. CORROSION ASSESSMENT

- Pipe wall thickness at elbow: 4.2 mm (original: 8.0 mm).
- Corrosion rate: 0.38 mm/year.
- External surface pitting observed near flange connection.
- Coating degradation: 40% of surface area affected.

4. SUMMARY

- Pump P-101 requires immediate maintenance attention.
- Seal replacement recommended within 48 hours.
- Vibration source to be investigated - suspect rotor imbalance or bearing degradation.
- Pipe section at CDU-E-103 elbow approaching minimum retirement thickness.

RECOMMENDATIONS & PRIORITY

PRIORITY 1 (IMMEDIATE · within 48 hours):

- Replace mechanical seal on Pump P-101.
- Verify seal alignment using laser alignment tool.
- Flush seal chamber and inspect seal cavity for debris.

PRIORITY 2 (URGENT · within 1 week):

- Perform detailed vibration spectrum analysis.
- Check rotor balance and bearing condition.
- Monitor bearing temperature continuously.

PRIORITY 3 (SCHEDULED · within 1 month):

- Schedule pipe wall thickness survey for CDU-E-103 section.
- Plan external coating repair program.
- Update corrosion monitoring records.

SAFETY NOTE:

This inspection report is provided for maintenance planning purposes.

All corrective actions must be reviewed and approved by the designated authority before execution.

Equipment shall not be returned to service without proper clearance.